Venkata Lakshmi Parimala Pasupuleti

GenAI Engineer · LLM Engineering · AI Agents · RAG · LangGraph · LangChain · LLMOps · Python · AWS · Azure · MLOps

+1-703-261-5410 | venkatalakshmiparimala@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala |
Virginia, USA — Open to Hybrid / Remote

PROFESSIONAL SUMMARY

GenAI Engineer with 9+ years of production engineering and 3+ years hands-on **LLM engineering** and **agentic AI** development — designing and deploying **AI agents**, **RAG architectures**, **LLM orchestration**, and **LLMOps** pipelines in production. Built **AgentProbe**: a 3-agent **LangGraph** runtime (Orchestrator → RAG Tracer → Auto Fix) with **multi-step reasoning**, **tool use**, inter-agent handoffs, and **multi-provider foundation model routing** (Anthropic, OpenAI, Groq, Ollama) — **LLMOps CI/CD** pipeline running automated **hallucination testing**, **grounding**, and **model evaluation** on every commit. Built **Leasel**: end-to-end **RAG knowledge base** pipeline (document ingestion, **chunking strategies**, **embedding pipelines**, **ChromaDB** vector retrieval, **reranking**) achieving **93.6% precision** on 1,240 enterprise documents. Enterprise AI pipeline across **1.6M BFSI financial records** (Python, PyTorch, OpenAI API) — eliminated 80% manual workload. Expert Python + SQL; AWS + Azure; Docker; **MLOps**; **data governance**. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

WORK EXPERIENCE

Magizhs Tech Solution
May 2025 – Aug 2025

AI/ML Engineer — GenAI · LLM Engineering · RAG Architectures · AI Agent Orchestration

- Designed and built a production **GenAI agent** system orchestrated as a **LangGraph** stateful graph (parse → plan → retrieve → validate → conditional routing) — **multi-step reasoning**, **tool use**, and **prompt engineering** strategies (chain-of-thought, structured output, memory mechanisms) translating natural-language requirements into structured operational plans; engineered **autonomous workflows** with conditional routing between compliant and non-compliant execution paths.
- Implemented a **RAG architecture** (document ingestion, **chunking strategies**, **embedding pipeline** over regulatory corpus, **ChromaDB**-compatible vector retrieval) grounding every AI decision in cited sources; designed **guardrails** and **compliance** validation layers — ensured **data quality**, **lineage**, and **observability** within the AI workflow before execution.

George Mason University
Aug 2024 – Aug 2025

Research Associate — ML Modeling · Model Evaluation · Data Pipelines · Analytics

- Built multi-source **ML** prediction models (Python, R) achieving **R^2 = 0.977** across 190+ countries; applied **model evaluation** frameworks, hypothesis testing, and cross-validation; surfaced anomalies using **Isolation Forest**; designed stakeholder dashboards (Tableau, ggplot2) improving comprehension **40%**; mentored **600+ students** in Python and database design.

Entrata
Jan 2017 – Aug 2024

Senior Software Engineer — Enterprise AI Pipelines · LLMOps · MLOps · Cloud · Data Engineering

- Shipped **enterprise GenAI pipeline** processing **1.6M BFSI financial records** (Python, PyTorch, OpenAI API) — custom neural network architecture for structured and unstructured data transformation; **data quality**, **lineage**, and **governance** controls; output validation and quality gates at launch; automated operational workflows replacing manual Excel operations; eliminated **80% of manual operational workload** with zero **compliance** violations.
- Built and maintained **REST APIs** and data pipelines (Python, PostgreSQL, RabbitMQ) for 100K+ monthly active users at 99.5% uptime; architected **AWS** cloud infrastructure (S3, EC2, RDS, Glue) with **serverless-compatible** pipeline design; led **microservices architecture** redesign (Docker — 50% scalability gain); optimized **200+ SQL queries** (30% performance gain); built **third-party integrations** (Salesforce, API connections).
- Built **CI/CD** pipelines (GitHub Actions) for **MLOps** and AI workloads including automated testing and **model versioning**; implemented **security** and **access control** (Libsodium encryption, 100% PHI compliance — zero security incidents); conducted 300+ peer code reviews; mentored 5+ engineers across 120+ Agile sprints.

PROJECTS

AgentProbe: LLMOps & GenAI Testing Toolkit — LangGraph · LLM-as-Judge · CI/CD Eval

Python · LangGraph · LangChain · FastAPI · ChromaDB · Anthropic · OpenAI · Groq · Ollama · React · Docker · GitHub Actions — github.com/venkatalakshmiparimala/AgentProbe

- Built a 3-agent **LangGraph** runtime (Orchestrator → RAG Tracer → Auto Fix) with **multi-step reasoning**, **tool use**, inter-agent state handoffs, and **multi-provider LLM** routing (Anthropic, OpenAI, Groq, Ollama) — 15-endpoint FastAPI backend; **ChromaDB vector database** with **embedding pipeline**, semantic retrieval, and agent observability tracing.
- Built 5-metric **LLM-as-judge model evaluation** engine (factual accuracy, **hallucination** testing, **grounding**, completeness, **compliance**) — **LLMOps CI/CD** (GitHub Actions) with automated **latency** and retrieval quality benchmarks on every commit; React dashboard with retrieval quality and performance monitoring; **Docker**-containerized, zero-cost local deployment.

Leasel: Production RAG Architecture — 93.6% Precision · Document Ingestion · Reranking

Python · LangGraph · RAG · ChromaDB · Scikit-learn · SQLite · reportlab — github.com/venkatalakshmiparimala/Lease_complaince_rag

- Built an enterprise **RAG architecture** pipeline (1,240 documents, 5 US states + federal standards) — **document ingestion**, **chunking strategies**, **embedding pipeline**, **ChromaDB vector database** with **metadata management** and **reranking**; **prompt engineering**, LLM-as-judge **hallucination testing** (pass@1 81.7%, pass@3 88.3%), deterministic rules engine, full **data lineage** and **governance**; achieved **93.6% precision, 87.9% F1** across 10 **compliance** rules; production-ready and fully auditable.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia · Machine learning, GenAI systems, RAG architectures, LLMOps, model evaluation, cloud AI, data governance

B.Tech Computer Science & Engineering — Jawaharlal Nehru Technological University, Anantapur — 2013 – 2017

TECHNICAL SKILLS

GenAI & LLM Engineering: GenAI, LLM Engineering, LLMOps, AI Agents, Multi-Step Reasoning, Tool Use, LangChain, LangGraph, LlamaIndex (familiar), Semantic Kernel (familiar), AutoGen (familiar), RAG Architectures, Prompt Engineering, Memory Mechanisms, Tool-Augmented Reasoning, Multi-Provider LLM Routing (OpenAI, Anthropic, Groq, Ollama, Azure OpenAI), Hallucination Testing, Grounding, Model Evaluation (LLM-as-Judge), Embedding Pipelines, Chunking Strategies, Document Ingestion, Metadata Management, Reranking, Agent Observability & Tracing

MLOps, Cloud & Data: MLOps, CI/CD (GitHub Actions), Docker, AWS (S3, EC2, RDS, Glue), Azure, GCP (familiar), Vector Databases (ChromaDB, FAISS-compatible, Pinecone familiar), Semantic Search, FastAPI (REST APIs), Microservices, PostgreSQL, MySQL, Python (expert), SQL (expert — 200+ production optimizations), R, JavaScript, Bash, Linux, Git, Agile/Scrum (120+ sprints)

ML & Analytics: PyTorch, Scikit-learn, XGBoost, Random Forest, Isolation Forest, BERT, spaCy, Hugging Face, NLP, Feature Engineering, Statistical Modeling, Data Governance, Data Lineage, Compliance Automation, Tableau, Power BI, ggplot2; BFSI domain (1.6M financial records: income limits, HUD payments, AMI threshold, rent processing)